```
printf("\n1.push\n 2.pop\n 3.Display\n 4.Palindrome\n  5.Exit\n");
printf("\nEnter your choice: ");
scanf("%d",&choice);
switch(choice)
{
case 1:push();
break;
case 2:pop();
break;
case 3:display();
break;
case 4:palindrome();
break;
case 5:return 0;
default: printf("end of execution\n");
}
}
```

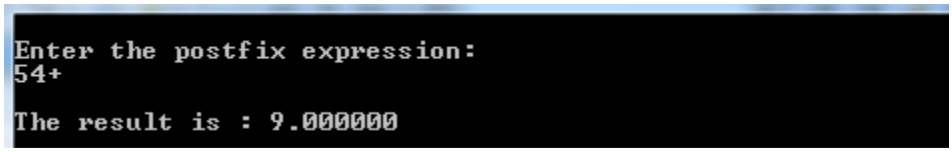
5. Develop a Program in C for the following Stack Applications

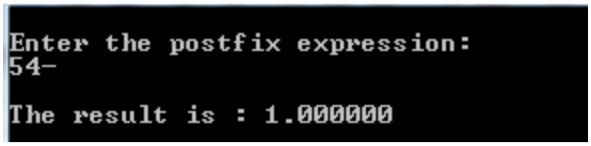
a. Evaluation of Suffix expression with single digit operands and operators: +, -, *, /, %, ^

b. Solving Tower of Hanoi problem with n disks.

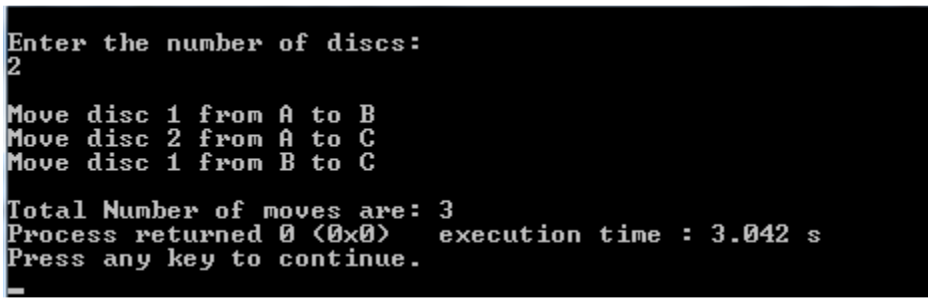
```
#include<stdio.h>
#include<conio.h>
#include<math.h>
#include<string.h>
double compute(char symbol, double op1, double op2)
{
switch(symbol)
{
case '+': return op1+op2;
case '-': return op1-op2;
case '*': return op1*op2;
case '/':return op1 / op2;
case '$':
case '^': return pow(op1,op2);
default: return 0;
}
```

```
}
int main()
{
double s[20], res, op1, op2;
int top, i;
char postfix[20], symbol;
printf("\nEnter the postfix expression:\n");
gets(postfix);
top=-1;
for(i=0; i<strlen(postfix); i++)
{
symbol = postfix[i];
if(isdigit(symbol))
s[++top] = symbol - '0';
else
{
op2 = s[top--];
op1 = s[top--];
res = compute(symbol, op1, op2);
s[++top] = res;
}
}
res = s[top--];
printf("\nThe result is : %f\n", res);
return 0;
}
```




**5b. Tower of Hanoi Problem**

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
void tower(int n, int source, int temp,int destination)
{
```

```
if(n == 0)
return;
tower(n-1, source, destination, temp);
printf("\nMove disc %d from %c to %c", n, source, destination);
tower(n-1, temp, source, destination);
}
int main()
{
int n;
printf("\nEnter the number of discs: \n");
scanf("%d", &n);
tower(n, 'A', 'B', 'C');
printf("\n\nTotal Number of moves are: %d", (int)pow(2,n)-1);
return 0;
}
```